

## F. Count the Numbers

time limit per test: 1 second

memory limit per test: 256 megabytes

**Forbidden Keywords for the Quiz:** open, file, creat(, fstream, thread, process, system(, exec(

You are given a sorted array  $a$  of  $n$  elements, and some queries. In each query, you are given a pair  $[x, y]$  and you have to count how many numbers  $a_i$  are there such that  $x \leq a_i \leq y$ . For example, if the array is  $[10, 20, 20, 45, 79]$  and you are given a query  $[20, 50]$ , then answer will be 3 because there are in total 3 numbers that's value is between 20 and 50.

### Input

The first line of the input contains  $n(1 \leq n \leq 10^5)$  and  $q(1 \leq q \leq 10^5)$  denoting the array size and the number of queries respectively. The next line will contain the array elements separated by space where  $1 \leq a_i \leq 10^9$  where  $i = 0, 1, 2, \dots, n - 1$ . Each of the next  $q$  lines will contain a pair  $[x, y]$  where  $1 \leq x \leq y \leq 10^9$ . See the sample input format for better understanding.

**Note1:** It is guaranteed that the given array is sorted in **non-decreasing** order.

**Note2:** It is also guaranteed that the queries are valid. Which means, for each query  $[x, y]$ ,  $x \leq y$ .

### Output

For each query  $[x, y]$ , output a single integer  $P$  denoting the number of elements in the array  $a$  such that  $x \leq a_i \leq y$ .

### Example

| input                                          | Copy |
|------------------------------------------------|------|
| <pre>5 3 10 20 20 45 79 20 50 5 45 1 100</pre> |      |
| output                                         | Copy |
| <pre>3 4 5</pre>                               |      |

### CSE221 Assignment 02 Summer 2026

Contest is running

22:34:32

Contestant



### → Submit?

Language: Java 21 64bit

Choose file:

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